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Task :- 2

Title

Feature Engineering, Model Optimization & Performance Comparison using California Housing Dataset

Objective

The objective of this task is to perform feature engineering, train multiple machine learning regression models, compare their performance using evaluation metrics, and identify the best-performing model for predicting house prices.

Dataset

The California Housing Dataset from scikit-learn was used. It contains housing-related features such as median income, house age, average rooms, average bedrooms, population, average occupancy, latitude, and longitude. The target variable is the median house value.

Methodology

Step 1: Data Loading

The California Housing Dataset was loaded using the scikit-learn library.

```
data = fetch_california_housing(as_frame=True)
```

```
X = data.data
```

```
y = data.target
```

```
print("Dataset Shape:", X.shape)
```

```
print(X.head())
```

Dataset Shape: (20640, 8)

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	\
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	

Longitude

0	-122.23
1	-122.22
2	-122.24
3	-122.25
4	-122.25

Step 2: Feature Scaling

The input features were standardized using StandardScaler to improve model performance.

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

Step 3: Data Splitting

The dataset was divided into training (80%) and testing (20%) sets using train_test_split.

```
x_train, x_test, y_train, y_test = train_test_split(
    x_scaled,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Samples:", len(x_train))
print("Testing Samples:", len(x_test))
```

```
Training Samples: 16512
Testing Samples: 4128
```

Step 4: Model Training

The following regression models were trained:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

```
models = {
    "Linear Regression": LinearRegression(),
    "Ridge Regression": Ridge(alpha=1.0),
    "Decision Tree": DecisionTreeRegressor(max_depth=5, random_state=42)
}

results = {}

for name, model in models.items():
    model.fit(x_train, y_train)

    predictions = model.predict(x_test)

    rmse = np.sqrt(mean_squared_error(y_test, predictions))
    r2 = r2_score(y_test, predictions)

    results[name] = {
        "RMSE": rmse,
        "R2 Score": r2
    }
```

Step 5: Model Evaluation

The models were evaluated using:

- Root Mean Squared Error (RMSE)
- R^2 Score

```
results_df = pd.DataFrame(results).T

print("\nModel Comparison")
print(results_df)
```

Model Comparison	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

Step 6: Performance Comparison

The evaluation results of all models were compared, and the model with the highest R^2 score and lowest RMSE was selected as the best-performing model.

```
best_model_name = results_df["R2 Score"].idxmax()

print("\nBest Model =", best_model_name)

best_model = models[best_model_name]
```

```
Best Model = Decision Tree
```

Results

The comparison table generated in the notebook shows the RMSE and R^2 Score for each model. Based on these metrics, the best-performing model was selected. An Actual vs Predicted scatter plot was also generated to visualize prediction accuracy.

```
y_pred = best_model.predict(X_test)

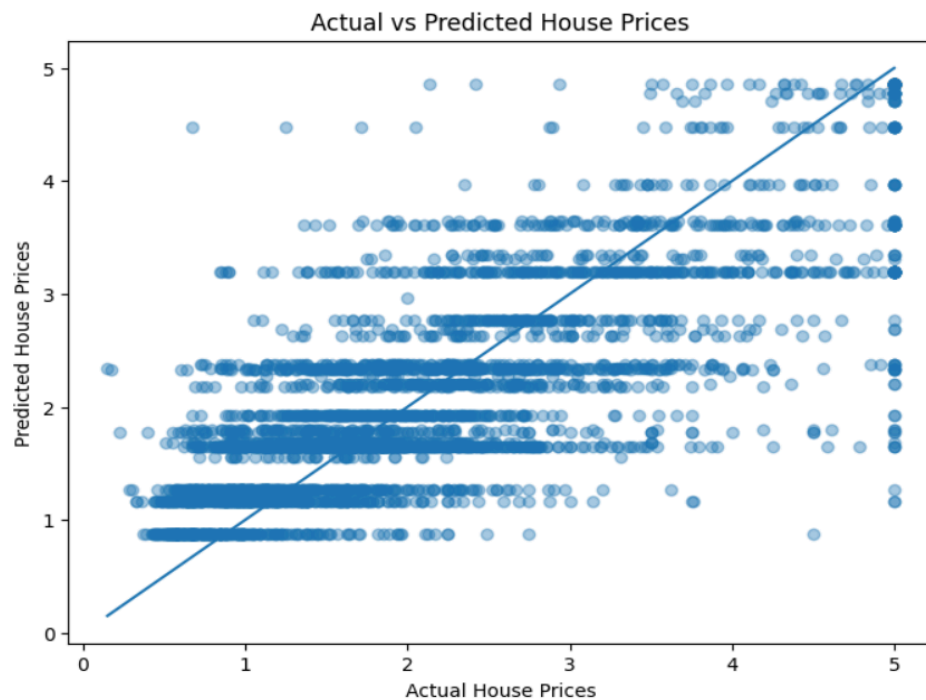
plt.figure(figsize=(8,6))

plt.scatter(y_test, y_pred, alpha=0.4)

plt.xlabel("Actual House Prices")
plt.ylabel("Predicted House Prices")
plt.title("Actual vs Predicted House Prices")

plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()]
)

plt.show()
```



```
import joblib

joblib.dump(best_model, "best_house_price_model.pkl")

print("\nModel Saved Successfully")
```

```
Model Saved Successfully
```

Conclusion

This task demonstrated how feature scaling and model comparison help improve prediction performance. Among the tested models, the model with the highest R^2 score was selected as the final model. The trained model was successfully saved for future use.